

Spatial heterogeneity and temporal dynamics of soil water tension in a mature Norway spruce stand

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Abstract:

In ecosystem research great effort is made in measuring soil water tension, because this is a critical calibration variable for modelling soil water fluxes. In this paper the spatial heterogeneity and temporal dynamics of soil tensions and their consequences for the determination of water fluxes are investigated. Studies were carried out at a Norway spruce stand in the Fichtelgebirge (NE Bavaria). Standard tensiometers were installed at three soil depths (20 each) on the whole experimental plot, as well as 45 microtensiometers as a dense grid in a small soil pit. Microtensiometry at the centimetre scale showed that, depending on rain intensity and initial soil water tension, even a soil without discernible macrostructure may show preferential water infiltration. At the stand scale the variability of soil hydraulic properties and tree root distribution causes substantial heterogeneity of soil water tension, as observed by standard tensiometers. A functional relationship between increasing spatial heterogeneity of tensiometer readings and increasing soil water tension was found, which was particularly pronounced after longer dry periods. Also at low soil water tension, where spatial heterogeneity was low, the calculation of water fluxes from tensiometer values was critical, owing to the fact that small differences in measuring soil water tension resulted in big differences in calculated water fluxes. At high soil water tension in summer the spatial heterogeneity of tensiometer readings was extremely high. At our experimental site, since 30% of the total rain in summer falls in events having a precipitation rate greater than 5 mm h^{-1} , preferential water and solute flow was an important phenomenon. We conclude that the validation of calculated water fluxes using measured soil water tension at the stand scale is not an appropriate tool, because of measurement difficulties, considerable spatial heterogeneity, especially in dry periods, and the great variability of soil hydraulic properties. © 1998 John Wiley & Sons, Ltd.

KEY WORDS tensiometry; spatial heterogeneity; temporal dynamics; water flux modelling

INTRODUCTION

Monitoring element fluxes with seepage in terrestrial ecosystems requires the installation of lysimeter cups, to record soil solution chemistry, and of tensiometers, for the registration of soil water tension, which is the critical reference variable for modelling water fluxes. Numerous papers have been published concerning ecosystem monitoring (for an overview see Harrison *et al.*, 1989). However, up to now little attention has been given to the implication of spatial variation of soil solution concentrations and soil water tensions for the determination of seepage fluxes.

Only in recent years the spatial variation of soil solution chemistry in forested ecosystems has been investigated in more detail. Grossmann and Kloss (1994) and Manderscheid and Matzner (1995a) looked at the spatial variability of soil solution chemistry in Norway spruce forests by installing more than 50 suction

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cups on their experimental plots and found that the coefficient of variation ranged from 12 to 298% depending on the ion considered. Although there are some detailed studies on the variability of tensiometer readings in agricultural soils, especially for the purposes of automatic irrigation (Saddiq *et al.*, 1985; Hendrickx *et al.*, 1990), only a few studies are available concerning the spatial heterogeneities of soil water tension in forest soils (Kreutzer, 1981; von Wilpert and Mies, 1991; Türk, 1992). However, they do not draw conclusions from these heterogeneities for tensiometer field installation or water flux calculation in a forest stand.

There are two distinct ways to treat field-scale effects of local soil variability. One is the attempt to identify local values of all relevant soil properties and then to simulate water fluxes on the basis of the resulting detailed distribution of tension (Manderscheid and Matzner, 1995b). This, however, is impractical over field scales (e.g. hundreds of metres), because the data and computational requirements are unreasonably large. In most applications people are more concerned with the large-scale (or spatially averaged) distribution of soil moisture. This has led to a number of investigators to seek for effective soil parameters that are able to aggregate the effects of local variability (Beven, 1995). The hope is that models based on these parameters will provide satisfactory predictions of large-scale flow features. However, the phenomenon of preferential flow, which includes macropore flow (Beven and Germann, 1982), fingering (Hillel and Baker, 1988) and funnelled flow (Kung, 1990), can lead to a rapid movement of water and solutes to great soil depth, although the occurrence of this phenomenon is very limited in space and time.

The variability of soil properties as well as the variability of water movement will cause spatial and temporal heterogeneities of soil water tension. The aim of our paper is to evaluate the magnitude and time dependence of these heterogeneities on different scales. The relevance of the observed heterogeneities for field installation and water flux calculations will be discussed.

MATERIALS AND METHODS

Site description

Tensiometer measurements are part of the standard ecosystem monitoring at the Waldstein experimental site. This site is located in the Fichtelgebirge (c. 780 m a.s.l), a chain of mountains in NE Bavaria (Germany) characterized by low mean annual temperature (5–6.5°C) and high annual precipitation (950–1050 mm). All installations are situated in the experimental plot 'Coulissenhieb' (area c. 2.7 ha). The age of the Norway spruce stand is about 140 years. The stem density amounts to 322 trees per ha. The soil is characterized as cambic podzol (FAO) derived from granite bedrock with a texture ranging from loamy sand to sandy loam. Further details are given in Manderscheid and Göttlein (1995).

Standard tensiometry

At the experimental plot 'Coulissenhieb' four subplots for element budget measurements were installed in 1992. Each of them consists of five samplers for bulk precipitation, as well as five suction lysimeters and five tensiometers in 20, 35 and 90 cm of soil depth each (see Figure 1). Tensiometers were constructed using a ceramic cup of length 5 cm and diameter 2 cm (P80 material from the 'Staatliche Porzellanmanufaktur', Berlin) and a Honeywell pressure transducer (type 136 PC). For automatic data registration a datalogger (Delta T Devices, Burwell, UK) was used. The recording was done hourly.

Microtensiometry

To install microtensiometers in a small soil pit, a Plexiglas plate with holes drilled in a 5 mm grid was fixed with long nails to one profile wall. This Plexiglas plate prevented the profile from damage and served as a mounting device for microtensiometer installation. Forty-five microtensiometers with an outer diameter of 1 mm (Göttlein *et al.*, 1996) were arranged in a 5 × 9 matrix with a horizontal and vertical spacing of 2 cm (Figure 2). In June and July 1994 tensiometer readings were recorded as 10 min averages using a Delta T datalogger.

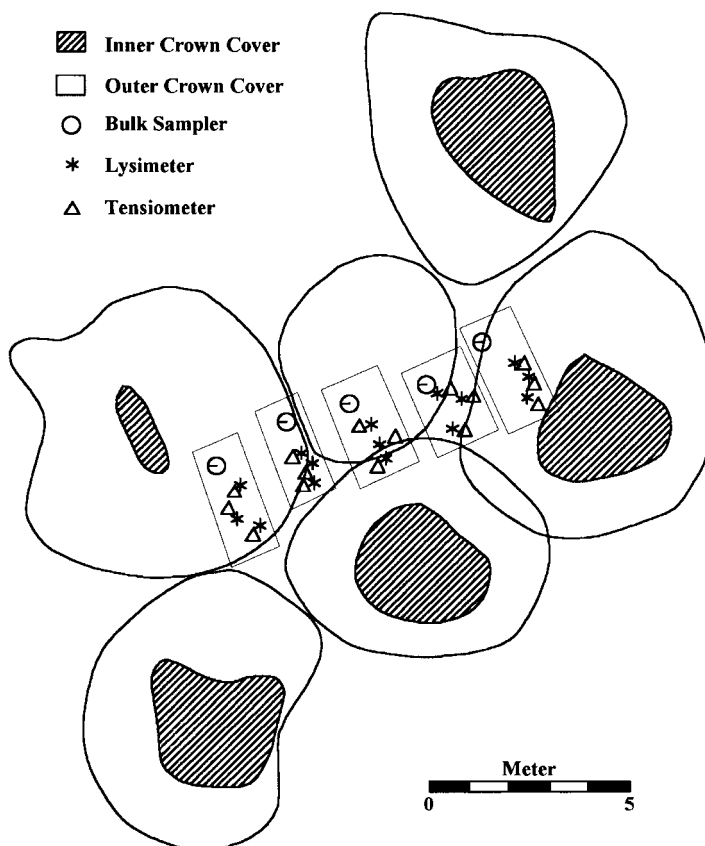


Figure 1. Location of the measuring equipment at one of the four measuring subplots at the experimental site 'Coulissenhieb'. Each subplot consists of five budget units, equipped with a bulk sampler and one suction lysimeter plus one tensiometer in 20, 35 and 90 cm soil depth each

RESULTS AND DISCUSSION

Heterogeneity of tensiometer readings on the micro-scale

Figure 3 shows the response of the uppermost (*c.* 4 cm depth) and lowest row (*c.* 20 cm depth) of micro-tensiometers at two consecutive rain events in summer 1994. At July 14th a heavy rain event with 10.4 mm of precipitation within 20 min occurred. One day later a comparable amount of precipitation (9.8 mm) fell within 1 h 50 min. For the first rain event all microtensiometers clearly reacted without any delay. For the second rain event only the uppermost row reacted immediately, whereas the lowest row of tensiometers showed only a slight and time-delayed decrease.

The picture becomes more complex when we look at high-resolution isoline maps of soil water tension (Figure 4). On July 14th at 17:00, just when the heavy rain began, soil water tension in the whole soil profile was about 160–200 hPa. Ten minutes after the beginning of the rain event tension at 20 cm depth decreased and at 17:20 was almost zero. At the same time, in the upper part of the soil profile, a suction up to 200 hPa was still measured. The rain event ended at 17:20. Subsequently the tension in the whole profile broke down without any observable wetting front, resulting in a more or less homogeneous distribution of soil water tension in the range 40–80 hPa at 19:00. The wetting of the soil, not from the top, but from some points in the lower part of the profile, is a strong indication that there are preferential flow paths behind the profile wall rapidly draining the water to greater soil depth.

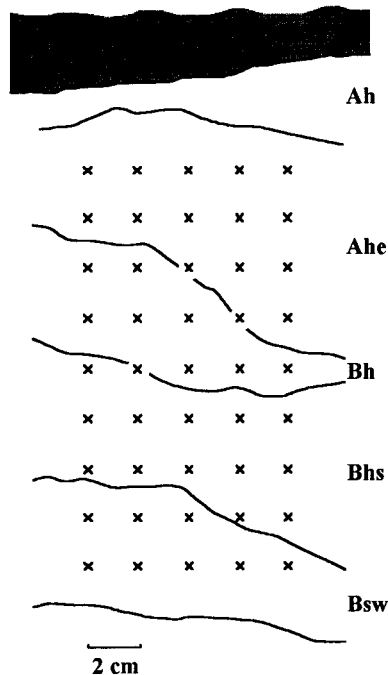


Figure 2. Installation scheme of 45 microtensiometers at the wall of a small soil pit at the experimental site 'Coulissenhieb'

The rain event on 15 July scarcely produced any preferential flow. Only at 5:00 in rows 4 and 6 two wet spots could be detected. In general, the soil profile was wetted from its upper part and an infiltration front moved down to deeper soil. This wetting front was very stable over time and could still be observed for more than 4 hours after the end of the rain event.

The heterogeneity of tensiometer readings within the soil profile during and after rain events is highly variable in space and time. Completely different spatial and temporal patterns could be observed depending on the occurrence of preferential flow. Preferential flow is important for the rapid transport of solutes to greater soil depth, which has been shown in particular for pesticides (Nordmeyer *et al.*, 1993). At the field scale, however, it is nearly impossible to predict or quantify preferential flow by common methods like tracer experiments (dyes, bromide), tensiometry or TDR (Demuth and Hiltbold, 1993). Ghodrati and Jury (1990) concluded that flow patterns are obviously three-dimensional, showing both lateral and vertical movement along tortuous paths. Our microtensiometer measurements showed that even in a soil without visible macropore structure water movement sometimes followed preferential flow paths and sometimes not, depending on the precipitation intensity and/or initial water content of the soil. Uneven moisture patterns when rewetting dry soils, ranging from sand to clay, have also been shown by Dekker and Ritsema (1996). Although several attempts were made to integrate the phenomenon of preferential flow into computer models (Germann, 1990; Gerke and van Genuchten, 1993), the exceptional variability of the soil system in time and space, hardly allows modelling of the real complexity of water fluxes on the stand scale.

Heterogeneity of tensiometer readings on the macro-scale

In Figure 5 the average time course for the year 1994 of the 20 tensiometers in 20 cm soil depth is shown (Figure 5a), as well as the standard deviation of tensiometer readings plotted against the average soil water tension (Figure 5b). During periods of high tension there was a high variability in tensiometer values, whereas at low tension in winter the variation between individual tensiometers was low. One interesting effect was that the 'history' of soil water saturation was important for the variation in tensiometer readings when

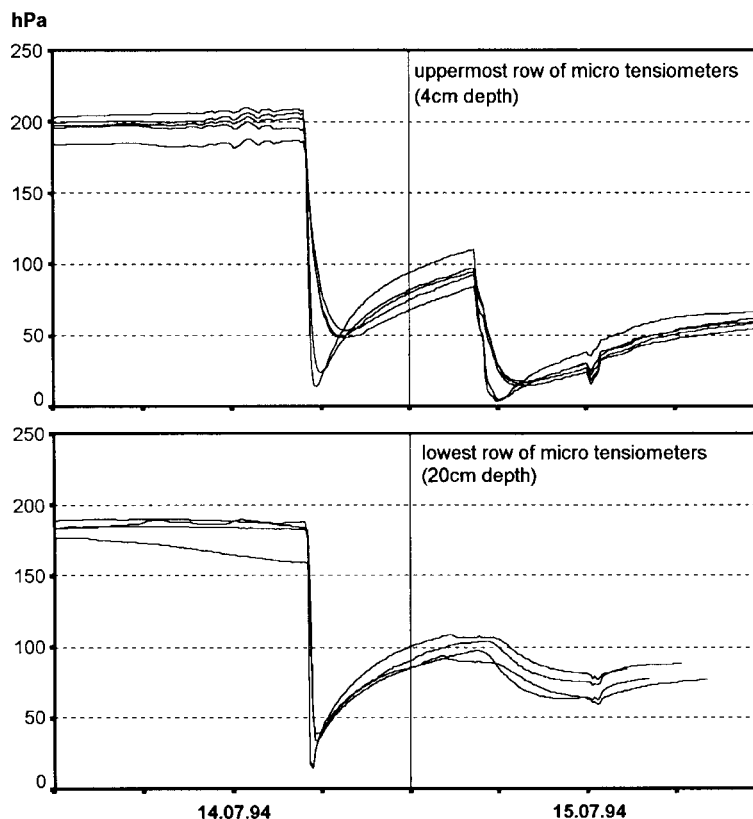


Figure 3. Response of the uppermost and lowermost row of microtensiometers for the two rain events in summer 1994

the soil was drying out. For the second dry period in July/August the standard deviation at comparable soil water tension was markedly higher than during the first dry period in June/July. This is because at the beginning of the first dry period the variation between individual tensiometers was relatively low. At the beginning of the second dry period, however, the variation between individual tensiometers was already relatively high, because rewetting of the soil after the first period of drought occurred non-uniformly.

During rewetting of dry soil the spatial heterogeneity of tensiometer readings often increased. This can be seen in Figure 5b, where in some cases standard deviations are clearly increasing with decreasing soil water tension, which occurred during rain events. There are two probable reasons for this effect. One reason can be deduced from Figure 1, where the canopy structure of the Norway spruce stand implies heterogeneous shading of the ground. A second reason may be the occurrence of preferential flow and macropores (Germann and Beven, 1985), where at some places water quickly moves to deeper mineral soil without having enough time for thoroughly rewetting the soil. In contrast, at places with low infiltration rates there will be enough time and water available to moisten the whole soil compartment completely. The increase of the standard deviation of soil water tension during drying of the soil is a well-known phenomenon (Mantoglou and Gelhar, 1987; Hendrickx *et al.*, 1994). In forest soils, however, this phenomenon is more pronounced than in agricultural soils, which are homogenized by ploughing and other mechanical soil tillage. Thus we found standard deviations to be more than threefold higher than those reported by Hendrickx *et al.* (1994) for an agricultural soil, especially under conditions when soil water tension exceeds 100 hPa. Not only is the lack of homogenization a reason for the high variation of tension in forest soils, but also the fact that trees, as perennial plants with high water demand and intensive rooting, cause an increase in the heterogeneity of tension when soil is becoming more dry (Kreutzer 1981; Grimmeisen *et al.*, 1986).

The diversity of the reaction of single tensiometers to rain events is presented in Figure 6. On July 14th the rain event leads to a rapid and simultaneous reaction of many tensiometers in 20, 35 and 90 cm soil depths. As shown in Figures 3 and 4 the simultaneous and rapid response of tensiometer readings in all soil depths indicates that this rain event caused preferential flow down to 90 cm soil depth and more in many places in our experimental site. However, some tensiometers showed only a slight and often time-delayed reaction. They were probably installed in soil compartments that are not closely connected to rapid flow paths. The rain event on July 15th could clearly be detected in 25 cm and, less intensely, in 35 cm soil depths. In 90 cm depth, with one exception, no rapid tensiometer response could be detected. Therefore, this rain event did not cause a relevant preferential flow to greater soil depth.

Consequences for field installation

To estimate the number of tensiometers necessary to register spatial averages of soil water tension Equation (1), according to Sachs (1984), was used. This equation calculates the minimal number of samples necessary to obtain the true average value of an underlying, normally distributed population at a given statistical certainty and precision. The precondition to reach the 'real' average tension at our experimental plot can be seen as a minimal requirement for the registration of spatial soil water tension heterogeneity:

$$n_{\bar{x}} = \left(\frac{Z_{\alpha}}{\bar{x} - \mu} \right)^2 \sigma^2 \quad (1)$$

with $n_{\bar{x}}$ is the minimal number of samples $Z_{\alpha} = 1.96$ for $\alpha = 0.05$ (95% certainty), $(\bar{x} - \mu)$ is the demanded precision and σ the standard deviation of the population.

The standard deviation (σ) was estimated conservatively in our case, with $\sigma = s$, where s is the standard deviation of the sample. The precision $(\bar{x} - \mu)$ was fixed at ± 15 hPa, which can be seen as a realistic requirement. The error deriving from the measuring equipment (pressure transducer, signal transmission, signal amplification and analogue–digital conversion, voltage supply) and from the calibration procedure of the pressure transducer can be estimated at about ± 10 hPa (Pröbstle *et al.*, 1991).

In Figure 5c the required number of replicates is plotted against average soil water tension. During winter and at tension lower than 100 hPa, ordinarily less than 50 tensiometers are necessary to register the real mean value of soil water tension at our experimental plot, with 95% statistical certainty and a precision of ± 15 hPa. For soil water tensions greater than 100 hPa, the required number of replicates drastically increases to more than 1000 at tensions higher than 300 hPa. As already seen in Figure 5b the second dry period in July/August had a higher standard deviation and thus would require a higher number of replicates, even at lower tensions. The clear distinction between summer and winter periods can also be seen in Table I,

Table I. Average number of replications required to obtain the 'true' average of soil water tension with 95% statistical certainty and a precision of ± 15 hPa depending on soil depth and different classes of tension

Tension (hPa)	Soil depth 20 cm		Soil depth 35 cm		Soil depth 90 cm	
	Oct–Mar	Apr–Sep	Oct–Mar	Apr–Sep	Oct–Mar	Apr–Sep
0–50	7	7	18	23	59	71
50–100	6	24	40	103	50	157
100–150	3	115	71	358	—	352
150–200	—	233	—	521	—	663
200–250	—	592	—	634	—	896
250–300	—	927	—	841	—	—
300–350	—	1103	—	996	—	—
350–400	—	1298	—	997	—	—
> 400	—	1427	—	—	—	—

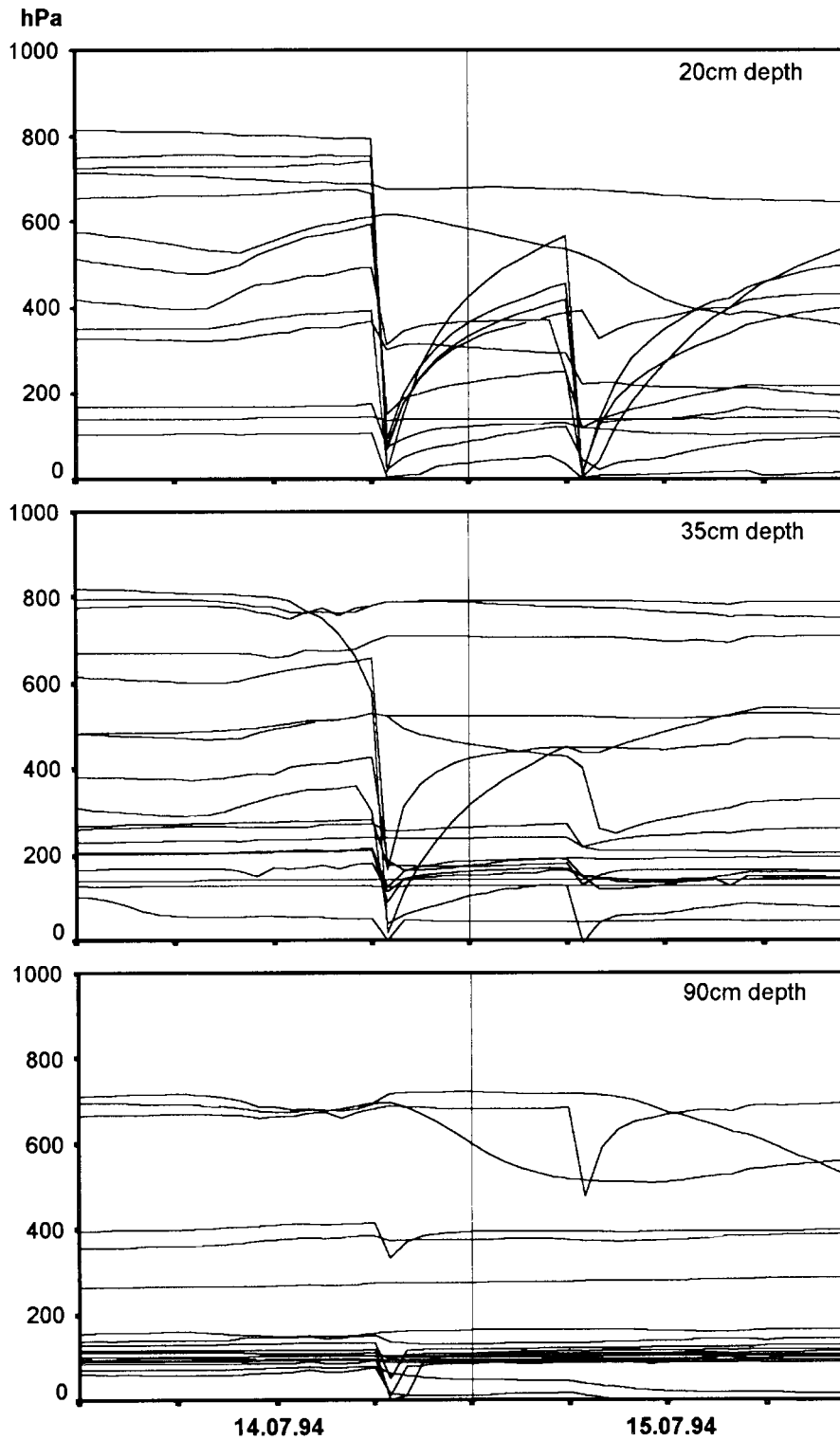


Figure 6. Response of individual standard tensiometer readings for the two rain events presented in Figure 3

where in the summer half-year the required number of replicates for all soil depths is higher. It is an interesting phenomenon that in deeper soil at low tension more tensiometer installations are required than in a 20 cm depth. This probably is owing to the fact that on our experimental plot two-layered soils are widespread, showing a change in soil texture at a depth of 40–70 cm (Manderscheid and Göttlein, 1995).

According to model calculations at our experimental plot more than 85% of total seepage movement occurs at tensions below 50 hPa. Taking this into account, the minimum number of tensiometers installed in 20 and 35 cm soil depths should be 10 and 20, respectively. In a 90 cm soil depth at least 70 tensiometers should be installed to cover representatively the spatial heterogeneity of soil water tension, a value that we have not reached in our field installation. With respect to the heterogeneity of soil water tension observed throughout the year, the installation of far more than 100 tensiometers per soil depth would be necessary, which is not practicable.

Models calculating water fluxes also need the hydraulic conductivity of the soil as an input variable. This input variable, however, is strongly affected by heterogeneity too. Jury (1985) and Nielsen *et al.* (1973) found that 576 samples are required to estimate saturated hydraulic conductivity with an accuracy of 10% at a 95% confidence level. At our experimental site a comparable variability of hydraulic conductivity can be observed (Table II), leading to a very high sample requirement (up to 450 samples for a precision of 10 cm d⁻¹), especially in the upper soil horizons. Such heterogeneities are known in groundwater hydrology as well, where in complex media the parameter $K(\Psi)$ is often regarded as a random field (Delhomme, 1979; Neuman, 1982).

Table II. Heterogeneity of hydraulic conductivity within a soil profile at the Waldstein experimental plot and estimation of samples required to reach a given precision [according to Equation (1)]

	Horizon	<i>n</i>	Saturated conductivity (cm d ⁻¹)		Required number of samples for precision	
			AVG	STD	10 cm d ⁻¹	5 cm d ⁻¹
Ahe	(0–9 cm)	14	60.7	108.6	453	1812
Bhs	(9–24 cm)	10	73.9	105.4	427	1707
Bsv	(24–38 cm)	10	136.6	92.4	328	1312
Bv	(38–54 cm)	16	61.2	38.2	56	224
(II) Cv	(> 54 cm)	15	14.9	18.6	13	53

n = number of samples; AVG = average; STD = standard deviation

Consequences for the calculation of water fluxes and the interpretation of tensiometer values

Often it is assumed that flow processes in the unsaturated zone can be modelled by deterministic means (Bear and Verrujit 1992), but there are many sources of heterogeneity that are difficult to deal with. Harr (1977) states that soil as a porous medium is made up of unknown geometries of discrete particles. This unknown feature can lead to a behaviour of the moving fluid that differs significantly from that expected from the usual deterministic models. There are also problems arising from heterogeneities in soil water tension and hydraulic conductivity and from preferential flow as discussed above. All these are reasons that the parameter variability for modelling flow processes is high. At tensions below 50 hPa the spatial heterogeneity is low and can be covered by an affordable number of tensiometers. Calculation of water fluxes in the range of low suction, however, is very sensitive to small differences in soil water tension and to errors in soil water tension measurement, because of the non-linear, steep increase of the hydraulic function in this range. In Figure 7 the cumulative water flux between 20 and 35 cm soil depth is calculated for individual budget units (compare Figure 1) on the basis of data for September 1994. Although the spatial heterogeneity of tensiometer readings is rather low, large differences in calculated water fluxes exist. The cumulative fluxes range from 9 to 35 mm of seepage. Taking the results of the budget units, an average infiltration of 20.0 mm is

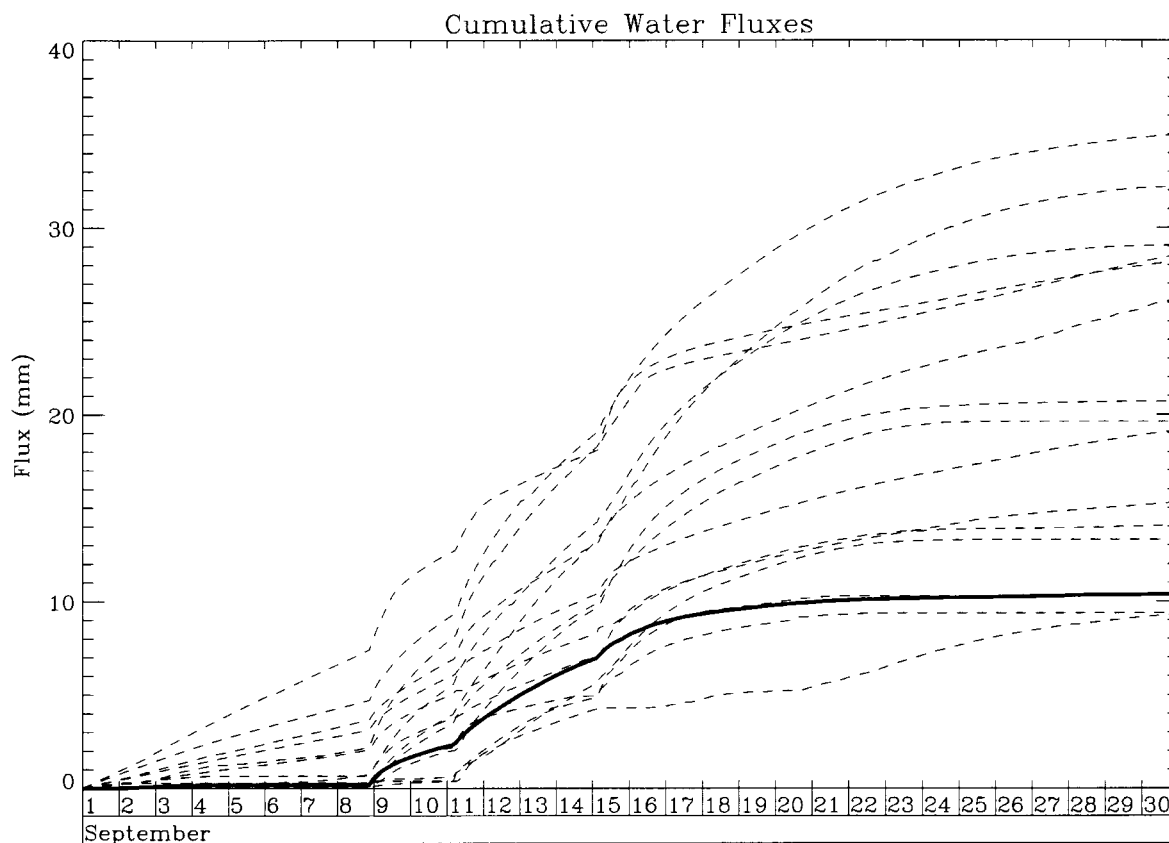


Figure 7. Cumulative water flux between 20 and 35 cm soil depth calculated for individual budget units (dashed lines) and averaged tensiometer values (solid line). Calculations were done using the Darcy equation, all with the same $K(\Psi)$ function

calculated. Owing to the high deviation between the individual budget units this average infiltration value is rather uncertain, with a standard deviation of 8.8 mm, equivalent to a coefficient of variation of 44%. Averaging the tensiometer values prior to flux calculation (bold line in Figure 7) results in a cumulative water flux of 10.4 mm, a value that is clearly different from the averaged individual fluxes. For all calculations we used the same $K(\Psi)$ function. However, one has to bear in mind that the hydraulic conductivity at a certain tensiometer position can never be known exactly and that there is a substantial spatial heterogeneity of hydraulic conductivity on our experimental plot (Table II). Thus, the differences in cumulative water fluxes between the individual budget units shown in Figure 7 are probably an underestimation of the diversity of fluxes really occurring on our experimental plot.

At high soil water tension, errors in tensiometer readings do not have severe consequences for the calculation of water fluxes. Because of the low hydraulic conductivity there is hardly any water movement under these conditions. In dry soils, however, the problem of preferential flow is more pronounced, resulting in a high spatial variability of soil tension during rewetting (Figure 5). Preferential flow paths can hardly be registered by tensiometry and thus water movement at such events is very difficult to estimate. Such heavy rain events are of significance at our experimental site (Table III), where 30% of the rain in the summer period falls at a rate of more than 5 mm h^{-1} . This means that about 30% of water and solute fluxes in summer, which are about 16% of the totals for the whole year, cannot be adequately handled by standard one-dimensional modelling procedures. Because the chemical concentrations in the preferentially moving water are also unknown it is nearly impossible to calculate the fluxes of dissolved matter caused by heavy rain events.

Table III. Distribution of rain amount and rain intensity at the experimental plot 'Coulissenhieb' in the summer and winter half-year (database 2/94–9/96)

Rain intensity (mm h ⁻¹)	Summer (%) (April–September) 571 mm	Winter (%) (October–March) 504 mm
0–1	24	41
1–2.5	26	40
2.5–5	20	15
> 5	30	4

To assess the risk of water stress for trees, the tensiometer readings at high soil matrix potential are particularly important (von Wilpert, 1991; Nilsson and Wiklund, 1992). When soil gets dry the heterogeneity of soil water tension increases to an extent that makes it impossible to determine the 'real' tension of the experimental plot. Another problem, which makes it difficult to determine whether trees suffer water stress, is uncertainty about their root distribution. Spruce and many other tree species are capable of forming sinker roots. Such roots can explore great soil depths (> 1 m) and improve water supply to the tree by reaching water resources in deeper mineral soil (Köstler *et al.*, 1968). Therefore, the estimation of water stress by tensiometry is not appropriate. Methods like predawn water potential or sap flow measurement should be more favourable, because they integrate over the whole root system of the tree. But these methods also have to face a higher between-tree variability when the water deficit increases (Lu *et al.*, 1995), which also makes it difficult to get representative measurements for a whole forest stand.

CONCLUSIONS

Soil water tension is a critical calibration variable for modelling soil water fluxes. At tensions below 50 hPa, where more than 85% of the water flux at our experimental site occurs, spatial heterogeneity is small and can be covered by an affordable number of tensiometer installations. The calculation of water fluxes in the range of low tensions is, however, very sensitive to small differences in soil water tension and therefore to errors in tensiometer measurement. Owing to the non-linear interdependencies of hydrological parameters, computing water fluxes with averaged $\Theta(\Psi)$ and $K(\Psi)$ relations cannot give a satisfactory solution to the problem. The calculation of water fluxes for representative subcompartments is no alternative because an unreasonably high number of subcompartments would be necessary to get a good estimation of the average water flux occurring at the experimental site.

At high soil water tension the spatial heterogeneity of tensiometer readings drastically increases, and the same soil profile may show preferential flow or normal rewetting, depending on rain intensity and initial soil water tension. In this case tensiometry cannot be used to register changes in soil water potential, because preferential and macropore flow cause a very heterogeneous distribution of water flow pathways. Criteria should be found to identify rain events causing preferential flow. Such events must be excluded from the normal Darcy-based calculation procedure.

As a result of the high uncertainty of water flux calculations in wet soils and the impossibility of dealing with preferential flow, the accuracy of Darcy-based modelling of water fluxes in forest stands is often unsatisfactory. The technical and computational expenditure necessary for conducting tensiometer measurements in most cases is not justified by the quality of the results obtained. More generalistic procedures, such as a water storage budget for the whole rooting zone, may be an alternative.

Another application of tensiometer measurements is the estimation of water stress conditions for trees, but for this purpose tensiometry is also a rather limited tool. Under dry conditions there is a very high spatial variation of soil matrix potential and it is nearly impossible to evaluate the water sources that trees can make

accessible. Beyond this, the measuring range of tensiometers is limited to about 900 hPa, a value at which water stress to plants is not yet severe.

ACKNOWLEDGEMENTS

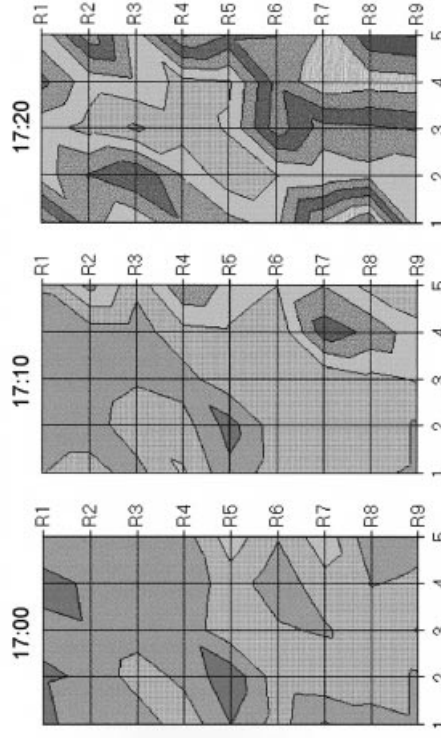
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Heavy Rain Event 14.7.94
17:00 to 17:20 h 10.4 mm



Rain Event 15.7.94
4:20 to 6:10 h 9.8 mm

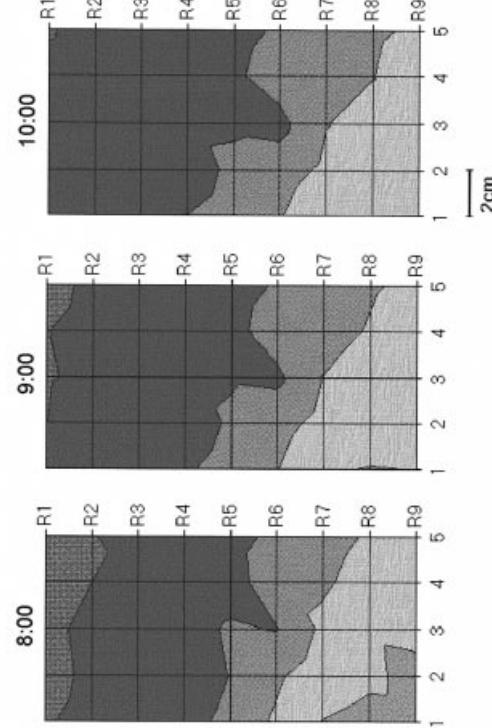
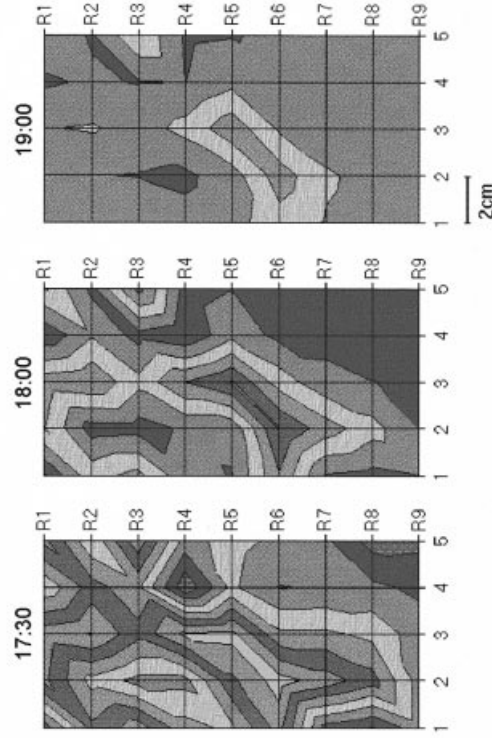
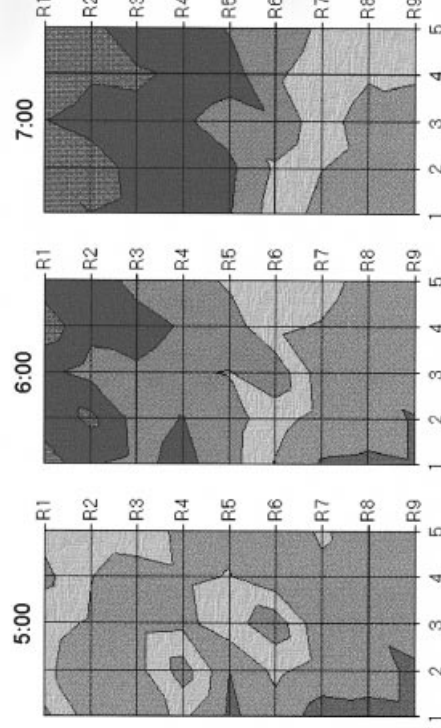


Figure 4. High resolution maps showing the changes of soil water tension during and after the rain events presented in Figure 3

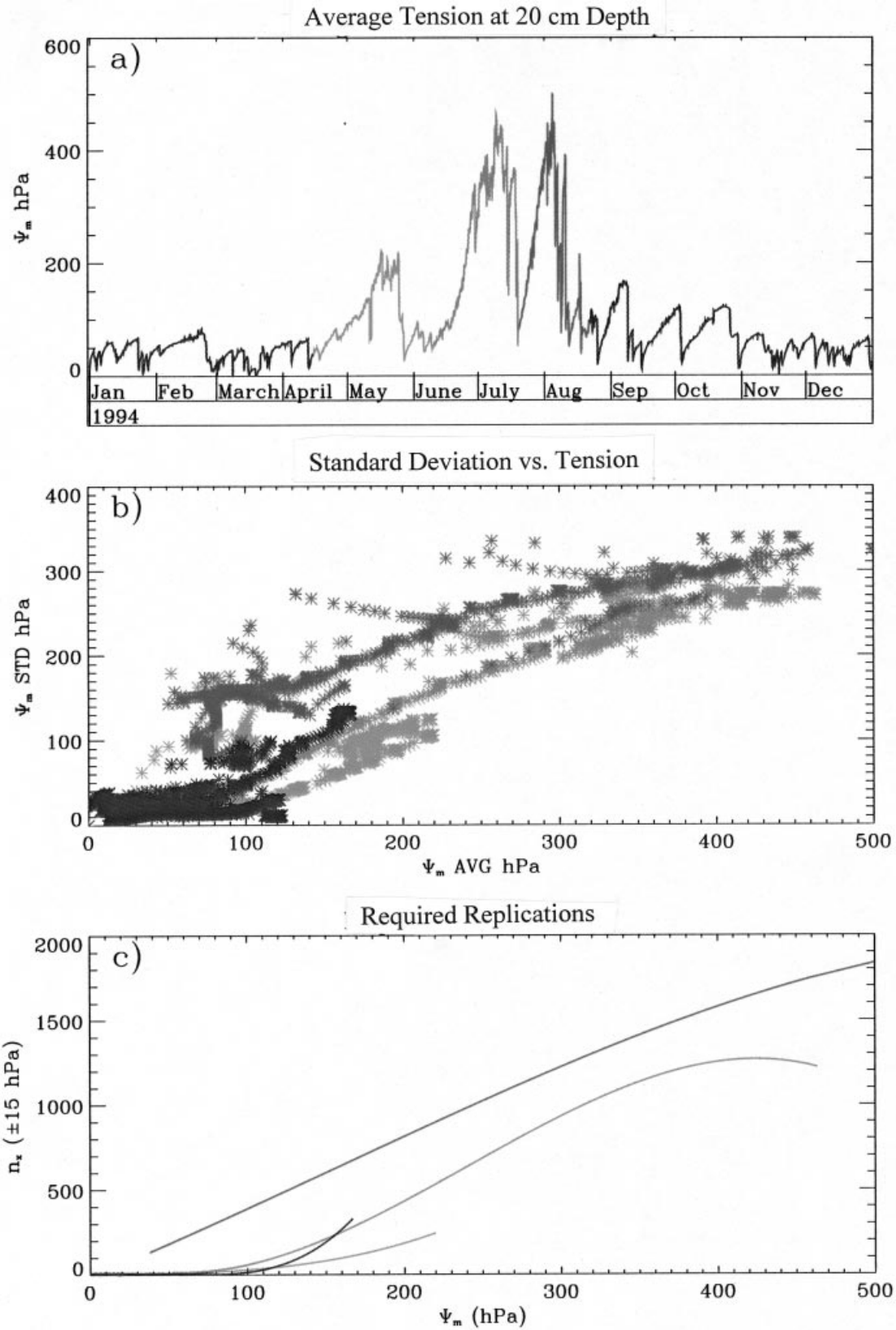


Figure 5. (a) Time course of average soil water tension for all tensiometers installed at 20 cm depth. Different colours indicate different characteristic periods. (b) Standard deviation of tensiometer readings depending on average soil water tension. (c) Estimation of the number of required tensiometer installations as calculated by Equation (1)